

Phase 9 CVRP Solver Evolution Report

Overview

- Condition count: 4.
- Best held-out condition: phase9_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Mean Novelty	Mean Complexity
phase9_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	42.59546	0.0	0.0
phase9_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	79.42246	0.0	0.0
phase9_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1365.54064	0.0	0.0
phase9_solver_evolution	solver_evolution	1.0	0.077593	0.077593	519.74634	0.540663	12.86333

Condition Notes

phase9_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 42.59546.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.

- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 79.42246.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1365.54064.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:
 - X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
 - X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_solver_evolution

- Execution mode: solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.103492.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.077593.
- Held-out runtime ms: 519.74634.
- Robustness dispersion across held-out structure families: 0.009168.
- Worst held-out instances:
 - X-n275-k28: feasible=True, penalized gap=0.100165, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.088987, errors=[]
 - X-n247-k50: feasible=True, penalized gap=0.077024, errors=[]

- X-n223-k34: feasible=True, penalized gap=0.075673, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.046118, errors=[]

Judge Note

****Summary of phase-9 CVRP Solver-Evolution Suite****

Aspect	phase9_baseline_nearest_neighbor_construction	phase9_baseline_clarke_wright_savings (Best Baseline)	phase9_baseline_regret_insertion_local_search	phase9_solver_evolution (Evolved Solver)
Feasibility	100% heldout & train	100% heldout & train	100% heldout & train	100% heldout & train
Objective Gap (Heldout)	0.2734	**0.0738**	0.4664	0.0776
Runtime (ms, Heldout)	42.6	79.4	1365.5	520.0
Robustness Dispersion	0.0028	0.0078	0.0869	0.0092
Complexity & Novelty	0 (fixed baseline)	0 (fixed baseline)	0 (fixed baseline)	Mean complexity ~12.9, novelty ~0.54

Interpretation

- ****Feasibility:**** All methods achieve full feasibility on held-out data.
- ****Objective Gap:**** The solver-evolution approach attains a heldout gap (0.0776) very close to the best baseline Clarke-Wright (0.0738), significantly better than Nearest Neighbor and Regret Insertion.
- ****Runtime:**** The evolved solver's runtime (~520 ms) is substantially greater than Clarke-Wright (~79 ms) but far less than Regret Insertion (~1365 ms).
- ****Robustness:**** Evolved solver robustness dispersion (0.0092) is slightly higher than Clarke-Wright (0.0078) but well below Regret Insertion.
- ****Complexity & Novelty:**** The evolved solver introduces meaningful novelty and complexity, showing effective evolutionary search.

Conclusion

The evolved solver:

- Maintains perfect feasibility.
- Achieves objective gaps nearly as good as the best baseline Clarke-Wright savings heuristic.
- Runs slower than Clarke-Wright but faster than Regret Insertion.
- Demonstrates stable performance (reasonable robustness).
- Does ****not clearly beat**** the best fixed baseline (Clarke-Wright) in objective gap or runtime but competes closely while adding novelty and complexity.

Thus, the evolution approach yields a ****competitive but not clearly superior**** solver compared to fixed baselines on held-out CVRP instances.